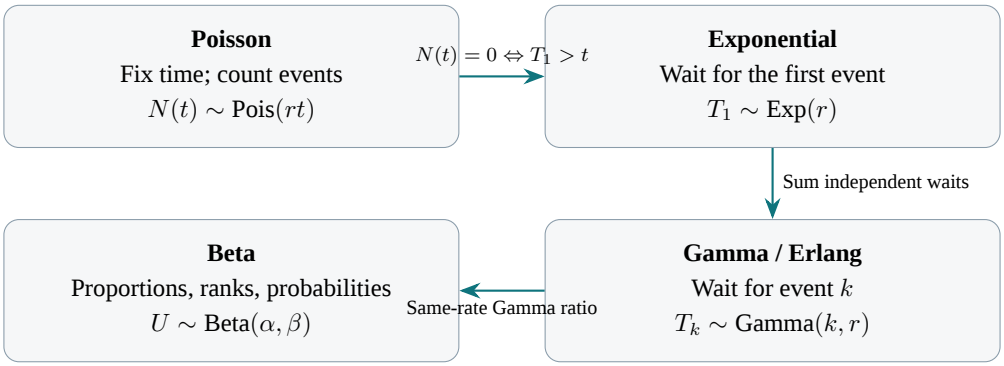
From Poisson Processes to Exponential, Gamma and Beta

A Unified Framework for Counts, Waiting Times, Order Statistics and Probability Inference



Core learning principle. First identify what is random, then identify the distribution. Every formula should be reconstructible from at least one of five ideas: **equivalent events, combinatorial counting, dimensional analysis, a probability kernel, or the law of total expectation**.
Memory devices. Three building blocks; slicing time; zero arrivals; length bias; cutting a cake; ghost votes; test points; springs and room to move.

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The four distributions in ten seconds

Distribution	Random quantity	Mean	Variance
Poisson(m)	Events in a fixed time window	$m = rt$	m
Exp(r)	Waiting time for event 1	$1/r$	$1/r^2$
Gamma(α, r)	Positive total / accumulated waits	α/r	α/r^2
Beta(α, β)	Proportion, rank, unknown probability	$\alpha/(\alpha+\beta)$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Start with four questions. Count arrivals in a fixed period: Poisson. Fix the number of arrivals and measure the time: Exponential / Gamma. Turn elapsed time into a fraction, or sort random points: Beta. Treat an unknown success probability itself as random: Beta conjugate inference.

Central question. Rather than memorize four isolated PDFs, ask: **is the random variable a count, a time, a location, or an unknown probability?** Reconstruct formulas using equivalent events, units, conditioning or normalization.

Learning sequence. (1) Poisson counts → (2) first wait (Exponential) → (3) k-th wait (Gamma) → (4) bus-waiting paradox → (5) cake cutting (Beta and order statistics) → (6) Gamma ratios → (7) Bayesian updating and prediction → (8) discrete waiting, integrals and interview problems.

Unified notation — essential before starting

- $r > 0$: arrival **rate per unit time**, with units $1/\text{time}$; e.g., $r = 0.1\,\text{min}^{-1}$. Never confuse the per-minute rate with the expected number per hour.
- $N(t)$: count on $[0, t]$; $N(t) \sim$ Poisson(rt). The parameter rt is the **expected count in that window** and is dimensionless.
- T_1: time from a fixed origin to the first event; T_k: time to event k; W: forward residual wait from a random observation time.
- Gamma consistently uses the **shape–rate** convention Gamma(α, r). Some books instead use the scale $\theta = 1/r$.
- Beta uses Beta(α, β), $\alpha, \beta > 0$, for random positions, proportions and unknown success probabilities on $[0, 1]$.

The Big Picture: Four Distributions, One Story

- **Fix time t; count events:** $N(t) \sim$ Poisson(rt), with $P(N(t) = k) = e^{-rt}(rt)^k/k!$. **Interpretation:** fix time, count how many.
- **Wait for one event:** $T_1 \sim$ Exp(r), with $P(T_1 > t) = e^{-rt}$. **Interpretation:** still waiting after $t \Leftrightarrow$ zero arrivals by t.
- **Wait for k events:** $T_k \sim$ Gamma(k, r) and $P(T_k \le t) = P(N(t) \ge k)$. **Interpretation:** the k-th event has occurred $\Leftrightarrow$ at least k arrivals.
- **Take a fraction of two independent Gamma totals:** $U = X/(X+Y) \sim$ Beta(α, β) when their rates match. **Interpretation:** absolute time → relative proportion.

- **Sort n independent uniform points:** $U_{(k)} \sim \text{Beta}(k, n-k+1)$. **Interpretation:** position threshold $\leftrightarrow$ number of points to its left.
- **Fix success probability p and wait for k successes:** $M \sim \text{NegBin}(k, p)$ and $E[M] = k/p$. **Interpretation:** fix the probability, count trials.

The two most important event equivalences

$$\boxed{\{T_1 > t\} = \{N(t) = 0\}}, \qquad \boxed{\{T_k \le t\} = \{N(t) \ge k\}}.$$

The whole book in one sentence. Poisson is a **count**; Exponential is **time to the first** event; Gamma is **time to event** k (integer k); Beta is a **proportion, position or probability**. The Gamma total-versus-fraction identity joins the two branches.

Poisson Processes: Fix Time, Count Arrivals

Distinguish the process from the distribution

Suppose events arrive randomly at a constant rate r. A **homogeneous Poisson process** satisfies: $N(0) = 0$; disjoint intervals have independent increments; equally long intervals have identically distributed increments; and in a short interval h, the probability of one arrival is $rh + o(h)$ while the probability of two or more is $o(h)$.

Consequently, for an interval $(s, s+t)$:

$$N(s+t) - N(s) \sim \text{Poisson}(rt), \qquad P(N(t) = k) = \frac{(rt)^k}{k!} e^{-rt}, \quad k = 0, 1, 2, \ldots$$

Never confuse the variables. r means "arrivals per minute"; rt means "expected arrivals over the next t minutes"; k is the actual integer count. If buses arrive on average three times per hour, the two-hour Poisson parameter is $3 \times 2 = 6$, not 3.

Why $e^{-m}m^k/k!$? The binomial limit

Cut a time interval of length t into n tiny slices. Each slice has approximately probability $p = rt/n$ of one arrival, so the total count is approximately $\text{Bin}(n, rt/n)$. Set $m = rt$, hold k fixed, and let $n \to \infty$:

$$P(N(t) = k) = \lim_{n\to\infty} \binom{n}{k} \left(\frac{m}{n}\right)^k \left(1 - \frac{m}{n}\right)^{n-k}.$$

$$\boxed{P(N(t) = k) = \frac{m^k e^{-m}}{k!}}.$$

How to remember the three pieces:

1. m^k **— the event engine.** To obtain k arrivals, each event contributes a factor related to the expected intensity m; hence the k-th power.
2. e^{-m} **— the silent baseline.** Almost all infinitesimal slices contain no event. Multiplying those "nothing happened" probabilities continuously gives e^{-m}, the probability of *zero arrivals throughout the window*. Think of it as the probability counterpart of continuous compounding, but decaying.
3. $k!$ **— removing ordering duplicates.** Counting k events in the same window ignores which arrival occurred first or second. The combination-counting limit leaves the factorial in the denominator.

These are *memory pictures*; the proof is the product of three limits: $\binom{n}{k} \sim n^k/k!$, $(m/n)^k = m^k/n^k$ and $(1 - m/n)^{n-k} \to e^{-m}$. The slice probability must be $m/n = rt/n$: it *decreases*, not increases, when n grows.

Mean, variance and "at least one"

$$E[N(t)] = rt, \qquad \text{Var}(N(t)) = rt, \qquad P(N(t) \geq 1) = 1 - e^{-rt}.$$

The moment-generating function (MGF) gives an independent check: $M_{N(t)}(z) = \exp[m(e^z - 1)]$, so $M'_{N(t)}(0) = m$ and $M''_{N(t)}(0) = m + m^2$, yielding variance m.

For Poisson, **mean = variance** refers to *counts in a fixed window*, not the waiting time's mean $1/r$ or variance $1/r^2$. The count is dimensionless, whereas r has units of inverse time: multiply by t first.

Interview Question: probability of a car in 5 minutes from 20 minutes

Given $P(N(20) \geq 1) = 609/625$, let q be the no-arrival probability in five minutes. Independent and stationary increments give:

$$q^4 = P(N(20) = 0) = 1 - \frac{609}{625} = \frac{16}{625}, \qquad q = \frac{2}{5}, \qquad \boxed{P(N(5) \geq 1) = \frac{3}{5}}.$$

Equivalently $q = e^{-5r}$ and $q^4 = e^{-20r}$. **Two assumptions matter:** arrivals in the four intervals are independent, and equal-length intervals have the same distribution. A constant *average* rate alone does not guarantee this; a fixed bus timetable violates independent increments.

Independence and joint events: Fréchet bounds

The Poisson multiplication rule, Gamma–Beta ratio identity and test-point proofs all rely on explicit **independence assumptions**. Marginal probabilities alone do not determine the joint probability. For two bonds with one-year default events A, B, suppose $P(A) = 0.5$, $P(B) = 0.3$. Then

$$\max(0, 0.5 + 0.3 - 1) \leq P(A \cap B) \leq \min(0.5, 0.3),$$

$$0.5 \leq P(A \cup B) = 0.8 - P(A \cap B) \leq 0.8.$$

These are **Fréchet bounds**. **Mutually exclusive $\neq$ independent:** if independent, $P(A \cap B) = 0.15$ and $P(A \cup B) = 0.65$; if exclusive, the intersection is zero and the union is 0.8. For binary default indicators, correlation ρ satisfies

$$P(A \cap B) = p_A p_B + \rho\sqrt{p_A(1 - p_A)p_B(1 - p_B)}.$$

The feasible range for this example is $-\sqrt{3/7} \leq \rho \leq \sqrt{3/7}$, *not* any value in $[-1, 1]$. In interviews, verify independence before multiplying probabilities.

Exponential Distribution: Wait for One Event

Derive the entire distribution from zero arrivals

Starting at any deterministic time, define $T_1 = \inf\{t \geq 0 : N(t) \geq 1\}$. Then

$$P(T_1 > t) = P(N(t) = 0) = e^{-rt}.$$

$$F_{T_1}(t) = 1 - e^{-rt}, \qquad f_{T_1}(t) = re^{-rt} \quad (t \geq 0).$$

Memory anchor: e^{-rt} is not a mysterious exponential function; it means "no event has occurred after t units of time." Differentiate $1 - e^{-rt}$ to get the PDF re^{-rt}.

Mean, variance and memorylessness

$$E[T_1] = \frac{1}{r}, \qquad \text{Var}(T_1) = \frac{1}{r^2}, \qquad \text{SD}(T_1) = \frac{1}{r}.$$

Use the tail-integral identity to rebuild the mean: $E[T_1] = \int_0^\infty P(T_1 > t)\, dt = \int_0^\infty e^{-rt}\, dt = 1/r$. The second moment is $E[T_1^2] = 2/r^2$ and therefore the variance is $1/r^2$.

$$\boxed{P(T_1 > s + t \mid T_1 > s) = \frac{e^{-r(s+t)}}{e^{-rs}} = e^{-rt}}.$$

Intuition for memorylessness. Waiting 20 minutes does not make an arrival more likely in the next five minutes *under a homogeneous Poisson model*. Its MGF is $M_T(z) = r/(r - z)$ for $z < r$, consistent with a Gamma distribution of shape 1. Memorylessness says *the conditional residual waiting-time distribution is unchanged*; independent increments say *counts in disjoint intervals are independent*. They are related, but **not literally the same definition**.

Dimensional analysis prevents confusing r with $1/r$

Suppose the arrival rate is $r = 6\,\text{h}^{-1}$. The expected count in one hour is six buses, while the **expected waiting time** is $1/6$ hour, or ten minutes. The **waiting-time variance** is $1/36$ hour^2. Poisson has **variance = mean** for counts; Exponential has **standard deviation = mean** for times. Variance always has squared units.

The exponential race: who arrives first?

For independent $E_i \sim \text{Exp}(r_i)$,

$$P(\min_i E_i > t) = \prod_i e^{-r_i t} = e^{-(\sum_i r_i)t}, \qquad \min_i E_i \sim \text{Exp}\left(\sum_i r_i\right).$$

The probability that clock j rings first is $P(E_j = \min_i E_i) = r_j / \sum_i r_i$. This is the foundation for exponential order statistics and the **addition of competing rates**.

Gamma: Wait for the k-th Event

From Exponential to Erlang, then to Gamma

Let successive interarrival times be $E_1, E_2, \ldots \overset{\text{iid}}{\sim} \text{Exp}(r)$. The k-th arrival time is $T_k = \sum_{j=1}^k E_j$ for a positive integer k. Hence $T_k \sim \text{Gamma}(k, r)$, called the **Erlang distribution** for integer shape.

When the shape α is not an integer, a Gamma distribution is still defined, but it cannot literally describe the arrival of a "2.7-th bus."

$$\boxed{f_X(x) = \frac{r^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-rx}, \quad x > 0, \quad \alpha, r > 0.}$$

The Gamma function is $\Gamma(\alpha) = \int_0^\infty z^{\alpha-1} e^{-z}\, dz$, satisfies $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$, and for a positive integer k satisfies $\Gamma(k) = (k - 1)!$. It supplies the **normalization**.

Density of the last arrival $\times$ exactly $k-1$ earlier arrivals

For "the k-th event happens in $(x, x+dx)$," there must be exactly $k-1$ arrivals in $[0,x]$, followed by one in the infinitesimal interval dx. Independent increments give

$$P(x < T_k < x + dx) \approx P(N(x) = k-1)r\,dx,$$

$$P(N(x) = k-1)r\,dx = e^{-rx}\frac{(rx)^{k-1}}{(k-1)!}r\,dx.$$

Therefore $f_{T_k}(x) = r^k x^{k-1} e^{-rx}/(k-1)!$.

Remember Gamma in four pieces:

1. x^{k-1}: exactly $k-1$ arrivals occurred *before* the last arrival.
2. e^{-rx}: the *silent baseline* formed by all tiny no-event intervals.
3. r^k: one arrival-rate factor per event; the density must also have units $1/\text{time}$.
4. $(k-1)! = \Gamma(k)$: the factorial/Gamma normalization.

Restriction. The event-count explanation is exact for integer k. For a noninteger shape α, Gamma is a continuous extension defined by the Gamma function, not a literal "arrival number α."

Compute the Gamma CDF using a Poisson tail

$$\boxed{P(T_k \le t) = P(N(t) \ge k) = 1 - e^{-rt}\sum_{j=0}^{k-1}\frac{(rt)^j}{j!}, \quad k \in \mathbb{N}.}$$

Conversely, $P(T_k > t) = P(N(t) \le k-1)$. In an interview this event equivalence is usually faster and clearer than directly integrating a Gamma PDF.

Mean, variance, MGF and parameterization

$$E[X] = \frac{\alpha}{r}, \qquad \text{Var}(X) = \frac{\alpha}{r^2}, \qquad M_X(s) = \left(\frac{r}{r-s}\right)^{\alpha} \quad (s < r).$$

For integer $\alpha = k$: add k independent exponential gaps: the mean is $k \times 1/r$, the variance is $k \times 1/r^2$, and the MGFs multiply. Gamma integration extends all three formulas to real $\alpha > 0$.

- **Shape–rate (this guide):** Gamma(α, r) with mean α/r and variance α/r^2.
- **Shape–scale (some books):** Gamma(α, θ) with mean $\alpha\theta$ and variance $\alpha\theta^2$.

Do not mix conventions. The formulas look identical at the notation level but the second parameters are reciprocals: $\theta = 1/r$.

Closure under sums of independent Gamma variables

For independent $X \sim \text{Gamma}(\alpha, r)$ and $Y \sim \text{Gamma}(\beta, r)$ with the **same rate**,

$$X + Y \sim \text{Gamma}(\alpha + \beta, r).$$

This follows immediately by multiplying MGFs. With different rates, the sum is *not generally Gamma* with shapes simply added.

Waiting at a Random Time: The Inspection Paradox

Sampling a gap versus landing in a gap

Suppose a Poisson bus process has average rate $r = 0.1/\text{min}$, or one bus per ten minutes. An *ordinary consecutive gap* $X \sim \text{Exp}(r)$ has mean $E[X] = 10$ minutes. But suppose you arrive at the station at a **stationary random time independent of the bus process**. Let A be the time since the last bus and W the time until the next. Then

$$A \sim \text{Exp}(r), \qquad W \sim \text{Exp}(r).$$

A and W are **independent** for the stationary Poisson process, because arrivals before and after the observation time correspond to independent increments. Therefore the length $L = A + W$ of the gap you occupy is $\text{Gamma}(2, r)$:

$$\boxed{E[A] = E[W] = 10 \text{ min}, \qquad E[L] = 20 \text{ min}.}$$

No contradiction. Randomly sampling a *gap* gives a mean of ten minutes. Randomly sampling a *time point* preferentially lands inside long gaps, so the occupied gap has mean 20. This is **length-biased sampling** or the inspection paradox.

Think of two gaps, one 1 minute and the other 19 minutes. Their ordinary average is 10 minutes; sampling a random time across the 20-minute period gives a mean occupied gap of $(1^2 + 19^2)/(1 + 19) = 18.1$ minutes. The longer gap acts like a much larger target.

The universal residual waiting-time formula for renewals

More generally, consider a **stationary renewal process** with IID nonnegative gaps X, mean $\mu = E[X] > 0$ and finite second moment $E[X^2]$. An observer arriving at a stationary random time lands in a **length-biased** interval:

$$E[L] = \frac{E[X^2]}{E[X]}, \qquad \boxed{E[W] = \frac{E[X^2]}{2E[X]} = \frac{\mu}{2} + \frac{\sigma^2}{2\mu}}.$$

A no-calculus derivation. Among realized intervals $X_1, \ldots, X_n$, the chance of landing in interval i is proportional to X_i. Therefore the mean occupied length is $\sum_i X_i^2 / \sum_i X_i \to E[X^2]/E[X]$. Conditional on length L, the observer's position is uniform, so the average remaining time is $L/2$.

Two ways to remember it: "**Each interval is weighted by its own length; then divide by two**." Or "**half the scheduled gap $\mu/2$ plus a variability penalty $\sigma^2/(2\mu)$**." Stationary sampling and the second-moment condition are essential; the formula cannot be applied indiscriminately to nonstationary processes.

- **Perfectly punctual (deterministic gaps):** $\mu = 10$ min, $\sigma^2 = 0$, hence $E[W] = 5$ min.
- **Poisson (exponential gaps):** $\mu = 10$ min, $\sigma^2 = 100\,\text{min}^2$, hence $E[W] = 10$ min.
- **Same mean, more irregular arrivals:** $\mu = 10$ min, larger σ^2, so $E[W] > 10$ min.

Beta: A Distribution of Positions and Proportions

Beta is not another waiting-time distribution

$$\boxed{f_U(u) = \frac{u^{\alpha-1}(1-u)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 < u < 1,}$$

$$B(\alpha,\beta)=\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\,du=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

The positive parameters α,β shape a random value between zero and one. $\text{Beta}(1,1)=\text{Uniform}(0,1)$. The density should *not* be interpreted literally as a probability of "$\alpha-1$ successes and $\beta-1$ failures"; that is only an analogy between the **kernel** and a likelihood.

Learn by recognizing probability kernels. First ignore the normalizing constant and identify the powers and exponential terms. Then use the Gamma/Beta integral to scale the area to 1. Beta kernel: $u^{\alpha-1}(1-u)^{\beta-1}$. Gamma kernel: $x^{\alpha-1}e^{-rx}$. Beta's normalizer is $1/B(\alpha,\beta)=\Gamma(\alpha+\beta)/[\Gamma(\alpha)\Gamma(\beta)]$. The shortcut "the Beta integral is always below 1" is *false for some positive parameters*; use the exact identity instead.

Beta mean and variance: relative balance and concentration

Set $S=\alpha+\beta$ and $\mu=\alpha/S$. Then

$$\boxed{E[U]=\frac{\alpha}{\alpha+\beta}=\mu},\qquad \boxed{\text{Var}(U)=\frac{\alpha\beta}{S^2(S+1)}=\frac{\mu(1-\mu)}{S+1}}.$$

Separate the roles. μ controls whether the center lies left or right. S controls concentration: with μ fixed, larger S means smaller variance. The "-1" in $\alpha-1$ and $\beta-1$ belongs to the PDF *exponents*; add 1 to each power when identifying a Beta distribution from an integral or likelihood.

Shape and limiting intuition

- $\alpha=\beta=1$: uniform. $\alpha=\beta>1$: concentrated in the middle. $0<\alpha=\beta<1$: more mass near the two ends.
- $\alpha>\beta$: mean exceeds $1/2$; $\alpha<\beta$: mean is below $1/2$.
- **Beta variance** $=\mu(1-\mu)/(S+1)$**:** looks like Bernoulli's $p(1-p)$, reduced by "total parameter mass + 1."

Order Statistics: Why Cutting a Cake Produces Beta

The k-th smallest point: $k-1$ left, $n-k$ right

Let $U_1,\dots,U_n\overset{\text{iid}}{\sim}\text{Uniform}(0,1)$, sorted as $U_{(1)}<\cdots<U_{(n)}$. For $U_{(k)}$ to lie near x, exactly $k-1$ points must lie to its left and $n-k$ to its right. Thus

$$f_{U_{(k)}}(x)=\frac{n!}{(k-1)!(n-k)!}x^{k-1}(1-x)^{n-k}.$$

$$U_{(k)}\sim\text{Beta}(k,n-k+1),\qquad 0<x<1.$$

Cake-cutting mnemonic. Place n cuts along $[0,1]$: they create $n+1$ cake pieces. The k-th cut has k **pieces** to its left (shape $\alpha=k$) and $n-k+1$ **pieces** to its right ($\beta=n-k+1$). **Parameters count pieces; density exponents count points.** That is why the PDF uses $\alpha-1$ and $\beta-1$.

Derive the mean without integrals: add one test point

Throw another independent $R \sim \text{Uniform}(0,1)$ onto the segment. Conditional on $U_{(k)} = x$, $P(R < x \mid U_{(k)} = x) = x$. The law of total expectation gives

$$E[U_{(k)}] = E[P(R < U_{(k)} \mid U_{(k)})] = P(R < U_{(k)}).$$

All $n+1$ IID points are symmetric. The chance the extra test point ranks among the first k is $k/(n+1)$, so

$$\boxed{E[U_{(k)}] = \frac{k}{n+1}}.$$

Why an expectation becomes a probability. Given that the k-th point is at x, the conditional probability of an independent test point lying left is exactly x. Taking expectations gives $E[E[\mathbf{1}_{\{R<U_{(k)}\}} \mid U_{(k)}]] = P(R < U_{(k)})$. Both independence and the conditioning step are essential: one cannot arbitrarily replace every expectation with a probability.

Add two test points for the second moment and variance

Independently add a red point R and blue point B. Given $U_{(k)} = x$, the chance both land to its left is x^2. Hence $E[U_{(k)}^2] = P(R, B < U_{(k)})$. In a random ordering of $n+2$ points, the two test points must occupy two of the first $k+1$ positions relative to the original k-th point:

$$E[U_{(k)}^2] = \frac{\binom{k+1}{2}}{\binom{n+2}{2}} = \frac{k(k+1)}{(n+1)(n+2)}.$$

Let $p_k = k/(n+1)$. Subtract the squared mean:

$$\boxed{\text{Var}(U_{(k)}) = \frac{k(n-k+1)}{(n+1)^2(n+2)} = \frac{p_k(1-p_k)}{n+2}}.$$

Intuition. A point needs room on *both* sides to move, so middle ranks tend to have larger variance at fixed n. Extreme ranks are constrained by the walls at zero and one. The denominator $n+2$ is the Beta total-parameter mass $(n+1)$ plus one.

Why are two ranked positions positively correlated?

For $i < j$, use two independently placed test points again:

$$E[U_{(i)}U_{(j)}] = P(R < U_{(i)},\ B < U_{(j)}) = \frac{i(j+1)}{(n+1)(n+2)}.$$

Do not skip the counting step. Regard the n original points as fixed in sorted order, leaving $n+1$ insertion slots. Red must be to the left of old point i, and blue to the left of old point j. Count the arrangements including the relative order of red and blue if they enter the same slot. Arrangements with red before blue contribute $ij - i(i-1)/2$; those with blue before red contribute $i(i+1)/2$. Their sum is exactly $i(j+1)$. There are $(n+1)(n+2)$ equally likely insertion permutations. Subtracting the product of the two means gives

$$\boxed{\text{Cov}(U_{(i)}, U_{(j)}) = \frac{i(n-j+1)}{(n+1)^2(n+2)} = \frac{p_i(1-p_j)}{n+2}}, \quad i \le j.$$

$$\boxed{\text{Corr}(U_{(i)}, U_{(j)}) = \sqrt{\frac{i(n-j+1)}{j(n-i+1)}}}, \quad i < j.$$

Intuition. If the i-th smallest point lies unusually far right, the j-th smallest often moves right with it. The numerator of the covariance is "space left of the earlier point p_i × space right of the later point $1 - p_j$." This is a *memory picture*; the exact value still follows from combinatorics.

Interview P32: correlation of the minimum and maximum of two uniforms

Let $n = 2$, $Y = \min(U_1, U_2) = U_{(1)}$ and $Z = \max(U_1, U_2) = U_{(2)}$. Then $E[Y] = 1/3$, $E[Z] = 2/3$ and $\text{Var}(Y) = \text{Var}(Z) = 1/18$. Sorting does not alter the product: $YZ = U_1U_2$, hence $E[YZ] = 1/4$. Therefore

$$\text{Cov}(Y, Z) = \frac{1}{4} - \frac{1}{3}\frac{2}{3} = \frac{1}{36}, \qquad \boxed{\text{Corr}(Y, Z) = \frac{1}{2}}.$$

In general, $\text{Corr}(U_{(1)}, U_{(n)}) = 1/n$.

Another example without joint densities. Suppose $X_1, X_2 \overset{\text{iid}}{\sim} N(0, 1)$; let $Y = \min(X_1, X_2)$ and $Z = \max(X_1, X_2)$. By symmetry $E[Z] = 1/\sqrt{\pi}$, $E[Y] = -1/\sqrt{\pi}$; since $YZ = X_1X_2$, $E[YZ] = 0$ and $\text{Cov}(Y, Z) = 1/\pi$.

Interview strategy for max/min: look for algebraic invariants first: $\min + \max = U_1 + U_2$, $\min \cdot \max = U_1U_2$, and $\max - \min = |U_1 - U_2|$. Only then consider a joint PDF.

General tools for other order-statistic questions

For IID random variables with arbitrary continuous parent CDF F,

$$P(X_{(n)} \le x) = F(x)^n, \qquad P(X_{(1)} > x) = [1 - F(x)]^n.$$

For continuous F, the probability integral transform makes $F(X_i) \sim \text{Uniform}(0, 1)$; hence $F(X_{(k)}) \sim \text{Beta}(k, n - k + 1)$. The *original* order statistic $X_{(k)}$ need not itself be Beta.

The $n + 1$ spacings formed by Uniform points, including both end spacings, are **identically distributed but not independent**; jointly they follow Dirichlet$(1, \ldots, 1)$. A sum of any m spacings follows Beta$(m, n+1-m)$. In particular, for $n \ge 2$,

$$U_{(n)} - U_{(1)} \sim \text{Beta}(n - 1, 2), \qquad E[U_{(n)} - U_{(1)}] = \frac{n-1}{n+1}.$$

Four general methods: (1) for two extreme values, use invariants (sum, product, absolute difference); (2) for Uniform positions and gaps, use Beta/Dirichlet; (3) for exponential samples, use memorylessness and added rates; (4) for general IID variables, use $F_{\text{max}} = F^n$ and $P(\min > x) = (1 - F)^n$, transforming through $F(X)$ if useful.

The exponential order-statistic race. If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(r)$, their successive ranked gaps are independent:

$$X_{(j)} - X_{(j-1)} \sim \text{Exp}((n - j + 1)r), \qquad X_{(0)} = 0.$$

Thus $E[X_{(k)}] = \sum_{j=1}^{k} 1/[(n-j+1)r]$ and $\text{Var}(X_{(k)}) = \sum_{j=1}^{k} 1/[(n-j+1)^2 r^2]$. Each time a timer rings, one of the n competing rates disappears; the remaining clocks restart probabilistically because of memorylessness.

Remember variance and covariance with springs and room to move

Room for one point to move. Visualize n sorted Uniform points as beads on a string trapped between walls at 0 and 1. For bead k, the expected room to the left is p_k; to the right it is $1 - p_k$. If either side has no room, motion is constrained. That suggests the variance numerator $p_k(1-p_k)$; the increasing number of beads adds the denominator $n + 2$.

Room for two points to move together. For $i < j$, when the left bead is pinned to the left wall or the right bead to the right wall, they have little freedom to move together. The two *outer slack regions* are p_i and $1 - p_j$, so the covariance numerator is $p_i(1 - p_j)$. This is a mnemonic, *not a physical derivation*; rigorous proofs use the joint order-statistic distribution or two-point counting.

Three formulas side by side:

$$E[U_{(k)}] = p_k, \qquad \text{Var}(U_{(k)}) = \frac{p_k(1-p_k)}{n+2}, \qquad \text{Cov}(U_{(i)}, U_{(j)}) = \frac{p_i(1-p_j)}{n+2} \ (i \le j).$$

Unify parameters and exponents. Beta's α, β can represent the *number of pieces* on either side of a cut (for integer ranks), or *effective vote weights* in Bayesian updating. The density exponents are $\alpha - 1, \beta - 1$. In particular Beta$(1, 1)$ has both exponents zero and is uniform. This parameterization needs no story about a "historical mistake."

Why Is a Ratio of Gamma Variables Beta?

Same-rate Gamma: independence of total and proportion

Let $X \sim \text{Gamma}(\alpha, r)$ and $Y \sim \text{Gamma}(\beta, r)$ be **independent with the same rate** r. Define

$$S = X + Y, \qquad U = \frac{X}{X+Y}.$$

Then

$$\boxed{S \sim \text{Gamma}(\alpha+\beta, r), \qquad U \sim \text{Beta}(\alpha, \beta), \qquad S \perp U.}$$

Intuition. S is the *total elapsed time*; U asks *what share belonged to phase one*. This is the Gamma–Beta algebra (sometimes introduced through a bank-versus-post-office example), the bridge between absolute quantities and random proportions.

See the Jacobian proof at a glance

Change variables via $x = su$ and $y = s(1-u)$, for $s > 0$ and $0 < u < 1$. The absolute Jacobian is s. The joint density becomes

$$f_{S,U}(s,u) = \frac{r^{\alpha+\beta} s^{\alpha+\beta-1} e^{-rs}}{\Gamma(\alpha)\Gamma(\beta)} u^{\alpha-1}(1-u)^{\beta-1}.$$

It factors into a density involving only s and a density involving only u, simultaneously proving the Beta normalization and **independence** $S \perp U$. If the Gamma rates differ, this result does not generally hold.

Beta as a Distribution for an Unknown Probability

Known success probability versus an uncertain probability

If the success probability p is given, the number of heads in n flips is $H \mid p \sim \text{Bin}(n, p)$. If even p is uncertain, encode uncertainty with a **prior distribution** over $[0, 1]$, such as $p \sim \text{Beta}(a, b)$.

After observing h heads and t tails, the **likelihood kernel** is $p^h(1-p)^t$. The binomial coefficient does not depend on p and can be omitted in a proportional likelihood. Bayes' rule gives

$$f(p \mid D) = \frac{P(D \mid p) f(p)}{P(D)} \propto p^{a-1+h}(1-p)^{b-1+t}.$$

Thus the classic **Beta–Binomial conjugate update** is

$$\boxed{p \mid D \sim \text{Beta}(a+h, b+t)}.$$

Ghost-vote mnemonic. For a uniform $\text{Beta}(1, 1)$ prior, imagine starting with "one ghost vote for heads and one for tails." Each observed head increases α by one; each tail increases β by one. **Parameters act like vote counts; PDF powers are parameter minus one.** This also resembles cake cutting, where the number of pieces is one more than the number of cuts. The metaphor does *not* mean that two trials were actually observed before data collection.

When does the extra $+1$ appear? Only when you start from $\text{Beta}(1, 1)$ do you obtain $\text{Beta}(h+1, t+1)$. Starting from $\text{Beta}(a, b)$, add just the observed h, t; **do not add another 1**. Treating a, b as literal past successes/failures is not a universally exact interpretation, especially when $a, b < 1$.

"God throws a dart": why cake cutting and posteriors both give Beta

Treat unknown p as a random $\text{Uniform}(0, 1)$ dart. Independently throw $n = h + t$ other Uniform test points. A test point left of p corresponds to heads and one right of p to tails. Conditional on exactly h points being left and t being right, the special dart p ranks $(h+1)$-st among the $n+1$ points. Hence

$$p \mid (h, t) \sim \text{Beta}(h+1, t+1).$$

This does *not* claim every Beta law originates from real coin tosses. It shows that, under a **uniform prior**, "the posterior of unknown p" and "the location of the $(h+1)$-st smallest Uniform point" have the same mathematical distribution.

Example: seven heads and three tails in ten flips

A uniform $\text{Beta}(1, 1)$ prior becomes $\text{Beta}(8, 4)$. The **posterior predictive probability** of heads next time is not simply $7/10$:

$$P(H_{11} = 1 \mid D) = E[p \mid D] = \frac{8}{12} = \frac{2}{3}.$$

The posterior mean is $8/(8+4)$, and the posterior variance is $(8 \cdot 4)/(12^2 \cdot 13) = 2/117$. The posterior is prior $\times$ likelihood, normalized; as sample size increases, the data typically receive greater weight.

Predict the number of successes in the next m trials: Beta–Binomial

Suppose $p \mid D \sim \text{Beta}(\alpha, \beta)$ and conditional on p the next m outcomes are independent, so $K \mid p \sim \text{Bin}(m, p)$. Integrate out the uncertain p:

$$P(K = x \mid D) = \int_0^1 \binom{m}{x} p^x (1-p)^{m-x} f(p \mid D)\, dp.$$

$$\boxed{P(K = x \mid D) = \binom{m}{x} \frac{B(\alpha + x, \beta + m - x)}{B(\alpha, \beta)}}.$$

For a contract that pays \$1 if there are *at least q heads in the next m flips*, the probability-based fair price (ignoring discounting and risk premia) is $\sum_{x=q}^{m} P(K = x \mid D)$.

For m **consecutive heads**, bypass integration through **sequential prediction and updating**:

$$\boxed{P(K = m \mid D) = \prod_{j=0}^{m-1} \frac{\alpha + j}{\alpha + \beta + j} = \frac{B(\alpha + m, \beta)}{B(\alpha, \beta)}}.$$

This is the Pólya-urn / sequential Bayesian predictive picture.

Symmetric posteriors and the boundary of a strict majority

If $\alpha = \beta$, then $P(K = x \mid D) = P(K = m - x \mid D)$. When m is **odd**, a strict majority of heads has probability $1/2$. When m is **even**,

$$P(K > m/2 \mid D) = \frac{1 - P(K = m/2 \mid D)}{2}.$$

A **strict majority** requires at least $\lfloor m/2 \rfloor + 1$ successes. For even m, using $\lceil m/2 \rceil$ as the cutoff incorrectly includes a tie.

Example: a trading strategy wins three times in a row

With a uniform prior, observing three successes and no failures produces a Beta$(4, 1)$ posterior. The probability under that posterior that the true win probability exceeds $1/2$ is

$$P(p > 1/2 \mid D) = 1 - \int_0^{1/2} 4p^3 \, dp = 1 - \frac{1}{16} = \frac{15}{16}.$$

Order-statistic intuition: the maximum of four Uniform points exceeds $1/2$ unless all four points lie in the left half. This is a **posterior probability conditional on the model**, not an unconditional guarantee about a real-world strategy.

Beta, Binomial and Negative Binomial: Positions Versus Counts

Fix n: is the k-th point to the left of p?

For n independent Uniform points, $U_{(k)} \le p$ exactly when at least k points lie in $[0, p]$. Each point independently falls to the left with probability p. Therefore

$$\boxed{P(U_{(k)} \le p) = \sum_{j=k}^{n} \binom{n}{j} p^j (1-p)^{n-j} = P(\text{Bin}(n, p) \ge k)}.$$

This expresses the Beta CDF as a binomial tail. Conversely,

$$\boxed{P(\text{Bin}(n, p) \le k - 1) = P(U_{(k)} > p) = 1 - I_p(k, n - k + 1)},$$

where $I_p(a, b) = B(a, b)^{-1} \int_0^p x^{a-1}(1-x)^{b-1} \, dx$ is the regularized incomplete Beta function.

Fix p: how many trials until the k-th success?

Each independent trial succeeds with probability p. Let M be the trial *on which the k-th success occurs*, **including that successful trial**. Under this convention the negative-binomial probability is

$$\boxed{P(M=n)=\binom{n-1}{k-1}p^k(1-p)^{n-k},\quad n\geq k}.$$

There are two ways to remember the mean. M is the sum of k independent Geom(p) random variables on $\{1,2,\ldots\}$, each with mean $1/p$; alternatively, regard each Uniform test point as a success if it falls in $[0,p]$. Thus

$$\boxed{E[M]=\frac{k}{p},\qquad \mathrm{Var}(M)=\frac{k(1-p)}{p^2}}.$$

Important distinction. Exponential describes *continuous-time waiting for one event*; Geometric describes *discrete-trial waiting for one success*, and is also memoryless. The equality $E[M]=k/p$ follows *exactly* from summing independent geometric waits; it is not proved merely by inverting the formula $E[U_{(k)}]=k/(n+1)$.

Which trader completes k wins first? A competition variation

Suppose Trader A has success probability p per trial and Trader B has success probability q, with independent trials both within and between traders. Each continues until reaching k successes. Let their required trial counts be M_A, M_B. The probability that **B finishes strictly before A** is

$$P(M_B<M_A)=\sum_{n=k}^{\infty}P(M_B=n)P(M_A>n),$$

$$P(M_B=n)=\binom{n-1}{k-1}q^k(1-q)^{n-k},$$

$$P(M_A>n)=\sum_{j=0}^{k-1}\binom{n}{j}p^j(1-p)^{n-j}.$$

The second term can be written as a Beta tail integral using the preceding identity. For large k, the CLT approximations $M_A\approx N(k/p,k(1-p)/p^2)$ and $M_B\approx N(k/q,k(1-q)/q^2)$ lead to a rough approximation from the normal distribution of their difference. **Caution:** an exponential rate suggested by a normal tail is an approximation, not an exact large-deviation exponent for the discrete competition event.

Gamma and Beta Integrals: Recognize the Kernel

Gamma kernels: from 0 to ∞

For $a>-1$ and $r>0$,

$$\boxed{\int_0^\infty x^a e^{-rx}\,dx=\frac{\Gamma(a+1)}{r^{a+1}}}.$$

Example: $\int_0^\infty x^4e^{-2x}\,dx=\Gamma(5)/2^5=4!/32=3/4$. Seeing x^a means **shape = exponent + 1**; seeing e^{-rx} means **rate =** r.

Beta kernels: from 0 to 1

For $a, b > -1$,

$$\boxed{\int_0^1 x^a(1-x)^b\,dx = B(a+1, b+1) = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)}}.$$

Example: $\int_0^1 x^3(1-x)^2\,dx = B(4,3) = 3!\,2!/6! = 1/60$. Add one to *both* exponents to obtain α and β; the denominator combines them in $\Gamma(\alpha+\beta)$.

Moments without integration: raise the shape parameters

$$E_{\text{Beta}(\alpha,\beta)}[U^m] = \frac{B(\alpha+m,\beta)}{B(\alpha,\beta)} = \prod_{j=0}^{m-1}\frac{\alpha+j}{\alpha+\beta+j},$$

$$E_{\text{Gamma}(\alpha,r)}[X^m] = \frac{\Gamma(\alpha+m)}{\Gamma(\alpha)r^m}, \qquad m \in \{0,1,2,\ldots\}.$$

Mnemonic. Gamma: an integer moment of order m raises the shape through m successive factors and divides by r^m. Beta: multiply m successive factors starting at α in the numerator, and m factors starting at $\alpha+\beta$ in the denominator.

Integrated Quant Interview Review

Problem statement → first reaction

- **Exactly k events in t minutes:** Poisson; $e^{-rt}(rt)^k/k!$.
- **At least one event, or none:** Poisson zero-event complement; $1-e^{-rt}$ or e^{-rt}.
- **How long to the next bus?** Exponential; $P(T>t) = e^{-rt}$ and mean $1/r$.
- **How long to bus number k?** Gamma / Poisson tail; $P(T_k \le t) = P(N(t) \ge k)$.
- **Random arrival time and surrounding gap:** length bias; $E[W] = E[X^2]/(2E[X])$.
- **k-th smallest of n Uniform draws:** Beta$(k, n-k+1)$.
- **Ratio of two Gamma variables:** Beta; check equal rates and independence.
- **Unknown success probability and new successes/failures:** Beta posterior; $(a,b) \mapsto (a+h, b+t)$.
- **Success count in the next batch:** Beta–Binomial; $\binom{m}{x}B(\alpha+x,\beta+m-x)/B(\alpha,\beta)$.
- **Wait for the k-th success:** Negative Binomial; $E[M] = k/p$.
- **Integrals $x^a e^{-rx}$ or $x^a(1-x)^b$:** Gamma / Beta kernel; add one to powers.

Four-distribution reference and unit checks

Distribution	Support / variable	Mean	Variance
Poisson(m), $m = rt$	Count $0, 1, 2, \ldots$	m	m
Exp(r)	Time $t \geq 0$	$1/r$	$1/r^2$
Gamma(α, r)	Time / positive total $x > 0$	α/r	α/r^2
Beta(α, β)	Proportion $0 < u < 1$	$\alpha/(\alpha+\beta)$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Ten high-frequency pitfalls

1. Whenever you see λ, check whether it denotes a **rate** or an **expected count over a window**; here the two are separated as r and rt.
2. Poisson(rt) counts events; Exp(r) and Gamma(k, r) measure elapsed time.
3. Gamma's second parameter might be a **rate** or a **scale**.
4. The "time to event k" interpretation of Gamma(shape k) needs a **positive integer** k.
5. Beta parameters are α, β; PDF exponents are $\alpha - 1, \beta - 1$.
6. A Uniform prior gives Beta$(h + 1, t + 1)$; a general Beta(a, b) prior gives Beta$(a + h, b + t)$.
7. Independent disjoint increments $\neq$ merely having a constant mean arrival rate; scheduled buses are not Poisson.
8. Randomly selecting a *gap* is different from randomly landing *inside a gap*.
9. Uniform order statistics are **not independent**; their spacings are also dependent, even though exchangeable.
10. A negative-binomial problem must specify whether M counts **all trials including the final success** or **failures before that success**.

Seven anchors you truly need to retain

1. **Zero arrivals:** $P(N(t) = 0) = e^{-rt}$.
2. **Equivalent events:** $T_k \leq t \iff N(t) \geq k$.
3. **Time units:** $E[\text{Exp}(r)] = 1/r$, Var $= 1/r^2$.
4. **Accumulated waits:** $\sum_{j=1}^{k} \text{Exp}(r) \sim \text{Gamma}(k, r)$ (independent terms).
5. **Ratio transformation:** $X/(X + Y) \sim \text{Beta}(\alpha, \beta)$ for independent, same-rate Gamma variables.
6. **Cake cutting:** the k-th smallest Uniform point has law Beta$(k, n - k + 1)$.
7. **Recognize the kernel:** Beta prior $\times$ Binomial likelihood $\Rightarrow$ add observed success/failure counts to the two parameters.

ABOUT THE AUTHOR

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I'm Mizuki Seika, a quantitative trader working at the intersection of mathematics, financial markets, and emerging technology; an explorer of Web3 and cryptocurrency; and, in a parallel universe that happens to be equally real, an independent idol and original music producer.

My academic path spans Carnegie Mellon University's Master of Science in Computational Finance (MSCF '28), Mathematics at Tsinghua University ('25), and an exchange year at the University of Oxford ('24). My interests range from quantitative research and systematic trading to decentralized finance and digital assets.

To me, markets and music have more in common than they appear to: both reward an ear for signals, a sense of structure, and the courage to build something original.

> *In finance, I turn complexity into models.*
> *In music, I turn it into melodies.*

This work is where I bring that same creative discipline to learning --- reconstructing knowledge until the pieces form a framework that actually makes sense.

Explore My Worlds

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Open-source projects and research

Spotify
Original music

Instagram
The creative side, on and off stage

Thank you for reading.